

Smart Logistics & Supply Chain Performance Project Report

Project Overview: End-to-end analytical pipeline taking a raw Smart Logistics & Supply Chain dataset from Kaggle, cleaning and structuring data using Python (Pandas) in Jupyter Notebook, storing and querying metrics with the help of MySQL, and presenting interactive insights via a Power BI Dashboard.

1. Executive Summary

This report evaluates performance metrics across 1,000 logistics tracking records recorded in the Smart Logistics & Supply Chain dataset. The analysis highlights key trends in asset utilization, inventory levels, logistics delays, and temporal operational patterns.

Key High-Level Metrics

- Total Logistics Records Analyzed: 1,000
- Total Unique Assets Tracked: 10 Trucks
- On-Time Deliveries: 434 records (56.60% Logistics Delay Rate)
- Average Asset Utilization: 79.60%
- Average Inventory Level: 297.92 units
- Average Demand Forecast: 199.28 units

2. Data Pipeline & Workflow Architecture

The analytics process followed a 4-step execution flow:

1. The smart_logistics_dataset.csv was loaded with 1,000 records and 16 initial columns
2. Pandas is used for data cleaning in Jupyter Notebook.
3. There are 263 missing values in the column "logistics_delay_reason" which were filled with the mode value of the column.

4. Standardized column headers to lower_snake_case to make the query process easy.
5. Created a new column for Asset Utilization category: Low Utilization (60-70%), Medium Utilization (71-85%) and High Utilization (86-100%).
6. Extracted month_name, month, week, day, day_of_week, hour and date from the column "timestamp".
7. Created Categories of purchase frequency: Low, Medium, High.
8. An Interactive Visual Dashboard (Power BI) summarizes asset performance, delay factors and inventory health.

3. SQL Analytical Insights & Query Results

3.1 Fleet Asset Utilization & Performance Ranking

Asset ID	Avg Asset Utilization (%)	Total Inventory Level	Delay Rate (%)	Performance Rank
Truck_7	80.67%	29,602	58.82%	1
Truck_1	80.52%	25,791	51.69%	2
Truck_8	80.24%	35,192	56.88%	3
Truck_2	80.16%	30,912	53.33%	4
Truck_3	80.10%	26,101	62.37%	5
Truck_9	79.94%	26,791	56.38%	6
Truck_10	79.58%	32,876	64.76%	7
Truck_4	79.06%	31,266	58.88%	8
Truck_6	79.03%	30,618	52.43%	9
Truck_5	76.60%	28,766	49.46%	10

3.2 Inventory Level Distribution & Stockout Risk

Inventory Level Category	Inventory Range	Record Share (%)	Delay Rate (%)
High	300+ units	49.50%	55.76%
Medium	151–300 units	38.50%	58.44%
Low	1–150 units	12.00%	54.17%

Stockout Risk Analysis: Identified critical instances where inventory level fell below demand forecast.

3.3 Temporal & Operational Bottlenecks

- **Peak Delay Month:** July has the highest total delay occurrences which is a total of 54 delays.
- **Peak Delay Day:** Tuesday has the highest average delay rate of 65.00%.
- **Peak Operational Hours:** Delays peaked at 12:00 PM having 33 delays, followed by 08:00 AM with 32 delays, and 10:00 AM with 31 delays.

4. Key Takeaways & Recommendations

1. **Service & Re-route High-Stress Trucks:** **Truck3, Truck7, and Truck_8** are over 80% utilization and facing >56% delays. Send them for maintenance and move them to clearer, lighter routes.
2. **Dynamic Inventory Buffers:** Automatically raise reorder points whenever forecasted demand exceeds the inventory level. This prevents stockouts.
3. **Avoid Rush Hours:** Block out non-urgent shipments during peak delay times (**8:00 AM – 12:00 PM**) and all **Tuesdays**. Shift these deliveries to afternoons or later in the week.